A4 — Finite Length and Biot Number for the Course Rod

PHYS 39 · Module 7 · **Kyle Chen** (individual) · Assignment A4 · November 2, 2026
Repository: https://github.com/hasanm720/Module1 (branch mainv2) · Git commit: **eee610fea41064f91ada01cb2564f00931145fc7**

Inputs are the measured rod geometry and sensor positions recorded before heating (docs/module_notes/module_07_rod_instrument.md), a literature range for k, and the provisional still-air coefficient h = 10 W m^{-2} K^{-1}. All values below are predictions; A4 does not use the Part III temperature-profile data.

Part 1 — Apply Fourier's law (Lienhard Problem 1.3)

Problem. A slab 1 cm thick carries a steady heat flux q = 5000 W/m² and its cold side is at –40 °C. Find the temperature rise to the hot side for seven materials and identify unreasonable cases.

Method. In steady one-dimensional conduction with no heat generation, the flux is the same at every plane, so Fourier's law integrates across the slab with constant k:

$$q = -k\frac{dT}{dx} \Rightarrow q = k\frac{\Delta T}{L} \Rightarrow \Delta T = \frac{qL}{k}$$

Units: (W m^{-2})(m)/(W m^{-1} K^{-1}) = K, so ΔT is in kelvin (equivalently °C for a difference). **Sample calculation (silver):** ΔT = (5000 W/m²)(0.01 m)/(427 W/m·K) = **0.117 K**, matching the textbook check value.

Material	k (W/m·K)	Source (Lienhard)	ΔT (K)	Hot side (°C)
Pure silver (99.99+%)	427	Table A.1, 20 °C	0.117	-39.9
Pure aluminum	237	Table A.1, 20 °C	0.211	-39.8
Mild steel (0.5% C)	54	Table A.1, 20 °C	0.926	-39.1
Ice	2.215	Table A.2, 0 °C	22.6	-17.4
Spruce	0.11	Table A.2, 20 °C	455	414.5
Insulation (85% magnesia)	0.067	Table A.2, 38 °C	746	706.3
Silica aerogel	0.024	Table A.2, 0 °C	2083	2043.3

Physical judgment. The metals need well under 1 K, and ice warms to about –17 °C, still frozen, so those four cases are reasonable. The last three are not:

• **Spruce:** the hot face would reach about 415 °C, far above the temperature at which wood chars and ignites. The slab could not survive this flux.
• **85% magnesia:** the hot face would reach about 706 °C, beyond the service temperature of this insulation, which is intended for moderate-temperature pipe lagging.
• **Silica aerogel:** a rise of about 2080 K puts the hot face above 2000 °C, past the melting point of silica; the aerogel structure would collapse long before.

These cases also break the calculation's own assumption: k was taken as constant, at one tabulated temperature, but over rises of hundreds of kelvin k changes and the material itself changes. The lesson is the one we apply to our TEC: a large heat flux can only be removed through a good conductor. If the path

from a heat source to its sink conducts poorly, the source temperature must climb until it fails, which is why the TEC is protected with a heat sink and an overtemperature shutdown.

Part 2 — Stationary finite-rod model (Lienhard §4.5)

Assumptions. Steady state; constant k and h; uniform cross section A and perimeter P; no internal heat generation; uniform ambient temperature T∞; and a transverse Biot number Bi⊥ = h(A/P)/k ■ 1, so each cross section has a single temperature T(x) (Lienhard eqn. 4.27; checked in Part 4).

Energy balance on a slice of length δx: heat conducted in at x equals heat conducted out at x + δx plus heat convected from the side area Pδx (eqn. 4.28):

$$-kA\frac{dT}{dx}\Big|_x = -kA\frac{dT}{dx}\Big|_{x+\delta x} + h(P\delta x)(T - T_\infty)$$

Divide by kA δx and let δx → 0. The difference quotient becomes the second derivative, and since T∞ is constant, d²T/dx² = d²(T – T∞)/dx² (eqns. 4.29–4.30). With θ = T – T∞:

$$\frac{d^2\theta}{dx^2} = m^2\theta, \qquad m^2 = \frac{hP}{kA} \quad [\mathrm{m}^{-2}]$$

Dimension check: hP/(kA) has units (W m^{-2} K^{-1})(m)/[(W m^{-1} K^{-1})(m²)] = m^{-2}, so m is an inverse length and mL is dimensionless. For a round rod, P/A = 4/d, so m² = 4h/(kd).

Dimensionless form. With Θ = θ/θ■ (θ■ = T■ – T∞) and ξ = x/L, the equation becomes d²Θ/dξ² = (mL)²Θ (eqn. 4.34), whose general solution is (eqn. 4.35):

$$\Theta = C_1 e^{mL\xi} + C_2 e^{-mL\xi}$$

Boundary conditions.
1. **Base, ξ = 0: Θ = 1** (fixed temperature, Dirichlet). Physically the base is held at T■; for our rod this is the temperature measured by the T1 thermistor at x = 0, not the TEC.
2. **Tip, ξ = 1: dΘ/dξ = 0** (insulated tip, Neumann). No heat crosses the end face, because its area A is small compared with the side area PL and its heat loss is neglected.

Substituting the general solution into both conditions gives C■ + C■ = 1 and C■e^{mL} – C■e^{-mL} = 0 (eqn. 4.37). Solving:

$$C_1 = \frac{e^{-mL}}{2\cosh mL}, \qquad C_2 = \frac{e^{mL}}{2\cosh mL}$$

Inserting the constants and using cosh z = (e^z + e^{-z})/2 gives the finite, insulated-tip profile (eqn. 4.41), its tip value (eqn. 4.45), and the base heat flow (eqns. 4.42–4.43):

$$\Theta = \frac{\cosh[mL(1-\xi)]}{\cosh mL} = \frac{\cosh[m(L-x)]}{\cosh mL}, \quad \Theta_{tip} = \frac{1}{\cosh mL}, \quad Q_0 = \sqrt{kAhP}\,\theta_0 \tanh mL$$

Part 3 — Semi-infinite limit and the provisional finite-length prediction

Lienhard Problem 4.12: the semi-infinite limit

Use the general solution θ = C■e^{mx} + C■e^{-mx} (eqn. 4.35, dimensional) with T(0) = T■ and T(L) = T∞, i.e. θ(0) = θ■ and θ(L) = 0:

$$C_1 + C_2 = \theta_0, \quad C_1 e^{mL} + C_2 e^{-mL} = 0 \Rightarrow \frac{\theta}{\theta_0} = \frac{\sinh[m(L-x)]}{\sinh mL}$$

For fixed x and L → ∞, sinh[m(L – x)]/sinh(mL) → e$^{m(L-x)}$/e^{mL}, so

$$\lim_{L\to\infty} \Theta = e^{-mx} \qquad \text{(eqn. 4.50)}$$

The insulated-tip profile reaches the same limit, since cosh[m(L – x)]/cosh(mL) → e^{-mx} when mL ■ 1. The semi-infinite model keeps the base condition but **loses the tip condition**; the rod is assumed so long that the tip has fallen to T∞. Lienhard recommends mL ■ 5 for temperatures and mL ■ 3 for Q■.

Example 4.8 and Problem 4.20: the cost of approximating the tip

Example 4.8 (2 cm aluminum rod, 8 cm long, k = 205 W/m·K, h ≈ h$_L$ = 120 W/m²K) has mL = 0.8656 and tip Biot number Bi$_{ax}$ = h$_L$ L/k = 0.0468. A convecting tip (eqn. 4.48) gives Θ$_{tip}$ = 0.6890; the insulated-tip approximation gives 0.7150, about 4% high. For the heat flow (Problem 4.20), the insulated tip gives Q■/(√(kAhP)θ■) = tanh mL = 0.6991, while eqn. 4.49 gives (Bi$_{ax}$/mL + tanh mL)/(1 + (Bi$_{ax}$/mL) tanh mL) = 0.7258. The approximation underestimates Q■ by **3.67%** (book: 3.67%).

For **our rod** with h$_L$ ≈ h = 10 W/m²K, Bi$_{ax}$ = hL/k ≈ 0.0168 and Bi$_{ax}$/mL ≈ 0.014, so the insulated tip predicts Θ$_{tip}$ = 0.542 compared with 0.536 for a convecting tip, a 1.2% difference. The insulated-tip model is adequate for our rod at this h.

Provisional prediction for the course rod

Inputs: d = 12.705 ± 0.025 mm; L = 283.2 ± 2 mm (T1 hole to tip); sensor positions x = 0.00, 50.30, 100.96, 150.32, 200.95 mm (± 0.5 mm) for T1, T2, T3, T4, T6; h = 10 W/m²K (provisional, still air); k = 166–172 W/m·K (6061-T6, Lienhard Table A.1, 0–100 °C, which spans our 20–45 °C operating range); error criterion ε$_T$ < 5% at every sensor.

$$m = \sqrt{\frac{4h}{kd}}, \qquad \varepsilon_T(x) = \frac{|\Theta_{finite} - \Theta_{\infty}|}{|\Theta_{finite}|}$$

m = 4.28–4.36 m^{-1} (decay length 1/m = 23.0–23.4 cm; the ±0.025 mm diameter uncertainty adds only ±0.1% to m). **mL = 1.21–1.23**, far below Lienhard's mL ■ 5.

Sensor	x (mm)	Θ finite	Θ semi-infinite	ε_T (%)	ε_T < 5%?
T1	0.00	1.000–1.000	1.000–1.000	0.0–0.0	yes
T2	50.30	0.838–0.842	0.803–0.806	4.1–4.2	yes
T3	100.96	0.715–0.722	0.644–0.649	9.9–10.1	no
T4	150.32	0.630–0.638	0.520–0.526	17.5–17.6	no
T6	200.95	0.572–0.581	0.417–0.423	27.1–27.2	no
tip	283.2	0.537–0.547	0.291–0.298	45.6–45.8	no

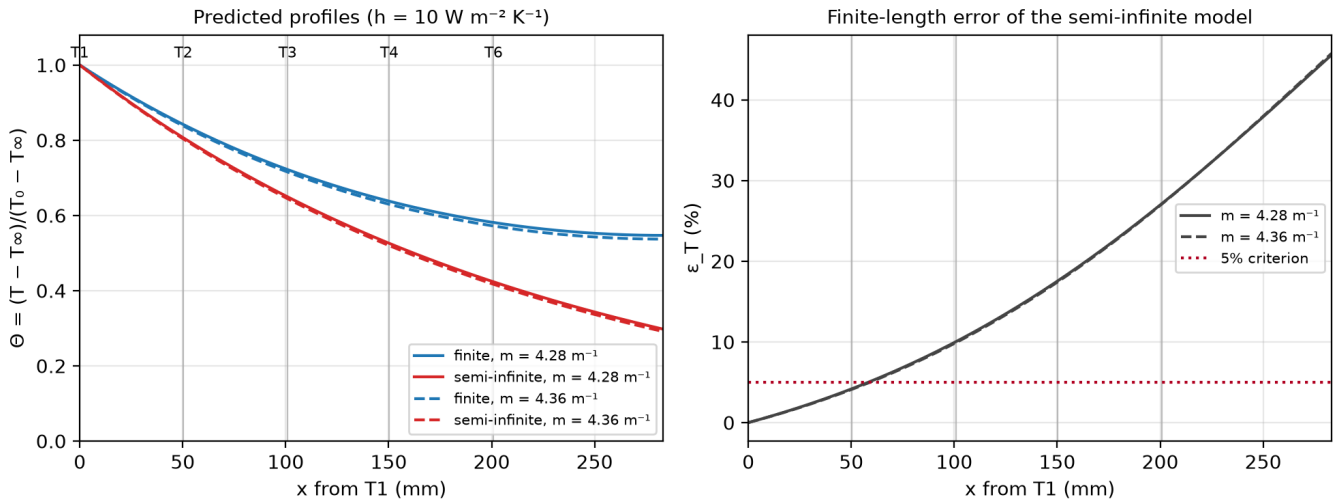


Figure 1. Predicted finite (insulated-tip) and semi-infinite profiles for the course rod at the two ends of the k range (left), and the finite-length error ε_T(x) (right). Vertical lines mark the five rod sensors.

Conclusion (length): the course rod is **not** adequately semi-infinite over the measured region. ε_T stays below 5% only at T1 and T2; it reaches about 10% at T3, 17% at T4 and 27% at T6, and the semi-infinite model would put the tip at $\Theta \approx 0.29$ instead of 0.54 ($\varepsilon_T \approx 46\%$). The stationary data should therefore be fitted with the finite-length model. This conclusion depends on the measured L and sensor positions, the provisional h, the k range, and the 5% criterion. Because $mL \propto \sqrt{h}$, a larger true h would make the rod look longer: mL would need to grow by a factor of about 4.1 (h about 17 times larger) to reach $mL \approx 5$.

Part 4 — Provisional one-dimensional (radius) prediction

The transverse Biot number compares conduction resistance across the radius with convection resistance at the surface (eqn. 4.27). For a circular rod A/P = R/2 = d/4:

$$Bi_{\perp} = \frac{h(A/P)}{k} = \frac{hR}{2k} = \frac{hd}{4k}$$

With h = 10 W/m²K, d = 12.705 mm and k = 166–172 W/m·K: **Bi⊥ = 1.85 × 10^{-4} – 1.91 × 10^{-4}** (dimensionless: (W m^{-2} K^{-1})(m)/(W m^{-1} K^{-1})).

Conclusion (radius): Bi⊥ is about 500 times smaller than the usual 0.1 threshold, so the temperature across any cross section is uniform to within a fraction of a percent of the local rod-to-air difference. **Each cross section is adequately represented by one temperature.** This is a separate question from Part 3: the rod is thin enough to be one-dimensional, but not long enough to be semi-infinite.

Application: a heat sink in still air vs forced airflow

Forced airflow thins the thermal boundary layer at the fin surface, so **forced flow has the larger convective coefficient h**; still air relies only on buoyancy-driven natural convection. Since m = √(hP/(kA)), the decay length scales as

$$\frac{1}{m} = \sqrt{\frac{kA}{hP}} \propto h^{-1/2}$$

so a larger h **shortens** the decay length: heat leaves the fin closer to its base, and the far part of the fin runs nearer the air temperature. For our rod, 1/m ≈ 23 cm at h = 10 W/m²K but only ≈ 7 cm at h = 100 W/m²K (a 10× increase in h shortens 1/m by √10 ≈ 3.2). The same heat sink therefore removes more heat with a fan, but a long fin gains less, because its outer part contributes little once 1/m is short. Bi⊥ also grows in proportion to h, yet even at 100 W/m²K it is only about 1.88 × 10^{-3}, so the one-dimensional assumption still holds.

Convection vs radiation (brief). Linearizing radiation about a mean surface temperature of about 32 °C gives h_{rad} = 4εσT■³ ≈ 6.5ε W/m²K. Bare machined aluminum has a low emissivity (roughly ε ≈ 0.05–0.1), so h_{rad} ≈ 0.3–0.6 W/m²K, small compared with h ≈ 10 W/m²K for natural convection. Radiation is a minor correction for this rod, but it would be comparable to natural convection for a black-anodized heat sink (ε ≈ 0.8–0.9), and negligible compared with forced airflow.

Sources and reproduction

Lienhard & Lienhard, *A Heat Transfer Textbook*, 6th ed.: Problem 1.3; §4.5 eqns. 4.27–4.51; Example 4.8; Problems 4.12, 4.20; Tables A.1–A.2. Rod geometry and sensor positions: docs/module_notes/module_07_rod_instrument.md. Build from the repository root (phys39a environment): /opt/anaconda3/envs/phys39a/bin/python code/module7/A4/build_a4.py